

Fasting plasma zeaxanthin response to *Fructus barbarum* L. (wolfberry; Kei Tze) in a food-based human supplementation trial.

Age-related macular degeneration (AMD) is a common disorder that causes irreversible loss of central vision. Increased intake of foods containing zeaxanthin may be effective in preventing AMD because the macula accumulates zeaxanthin and lutein, oxygenated carotenoids with antioxidant and blue light-absorbing properties. *Lycium barbarum* L. is a small red berry known as *Fructus lycii* and wolfberry in the West, and Kei Tze and Gou Qi Zi in Asia. Wolfberry is rich in zeaxanthin dipalmitate, and is valued in Chinese culture for being good for vision. The aim of this study, which was a single-blinded, placebo-controlled, human intervention trial of parallel design, was to provide data on how fasting plasma zeaxanthin concentration changes as a result of dietary supplementation with whole wolfberries. Fasting blood was collected from healthy, consenting subjects; fourteen subjects took 15 g/d wolfberry (estimated to contain almost 3 mg zeaxanthin) for 28 d. Repeat fasting blood was collected on day 29. Age- and sex-matched controls (n 13) took no wolfberry. Responses in the two groups were compared using the Mann-Whitney test. After supplementation, plasma zeaxanthin increased 2.5-fold: mean values on day 1 and 29 were 0.038 (sem 0.003) and 0.096 (sem 0.009) micromol/l ($P < 0.01$), respectively, for the supplementation group; and 0.038 (sem 0.003) and 0.043 (sem 0.003) micromol/l ($P > 0.05$), respectively, for the control group. This human supplementation trial shows that zeaxanthin in whole wolfberries is bioavailable and that intake of a modest daily amount markedly increases fasting plasma zeaxanthin levels. These new data will support further study of dietary strategies to maintain macular pigment density.